Circular Economy Understanding

OBJECTIVES

- Increase the ability to identify problems, characterize opportunities, formulate models and obtain solutions, emphasizing aspects of (i) sustainability; (ii) innovation; (iii) protection of privacy; and (iv) business analysis with decisions adjusted to risk, considering a hierarchy of portfolios, programs, projects/extensions and operations.
- Develop teamwork initially oriented to the understanding of Dutch PBL R10 model (Refuse, Rethink, Reduce, Reuse, Repair, Refurbish, Remanufacture, Repurpose, Recycle, and Recover Energy), followed by an adaptation to the P3RD model for Solid Waste/Residue Systems (P-Prevent, R1-Reduce, R2-Reuse, R3-Recycling, D-Disposal in Biodigester Sanitary Landfills), more suitable to developing countries.
- Describe the P3RD System Components in terms of Function/Structure/Process/Context, for systems and products developed in the light of the systems approach and oriented to sustainability (S), innovation (I), privacy (P), business analysis with decisions adjusted to risk (A), considering the P3RD model framework: (i) DM-Digital Mobilizer: SRSDM.SIPA-CC; (ii) R2-Reuse of solid waste via waste pickers' cooperatives: SRSR2.SIPA-CC; (iii) R3-Recycling of solid waste via waste pickers' cooperatives: SRSR3.SIPA-CC); (iv) DS-Disposal of solid waste into Biodigester Sanitary Landfills: SRSDS.SIPA; and (v) Products oriented to P-Prevention and R1-Reduction: SRSPR1.SIPA.
- Consolidate the understanding of relevant topics in systems projects: Active Learning (PBL Approach, Flipped Classroom, Teamwork); Project Management; Bibliometric Searching.

CIRCULAR ECONOMY: DEVELOPED AND DEVELOPING COUNTRIES PERSPECTIVES

The Dutch government has implemented a circular economy system with a vision extending up to 2050, structured around the R10 framework (**Refuse**, **Rethink**, **Reduce**, **Reuse**, **Repair**, **Refurbish**, **Remanufacture**, **Repurpose**, **Recycle**, and **Recover Energy**). This model, which practically excludes **landfills** from the circular economy cycle, is particularly well-suited for developed countries, like the Netherlands. In developing countries where both, the construction of new **Sanitary Landfills** working as **Biodigesters** and the required social inclusion of the people that now make a living as waste pickers in the thousands of open landfills now being closed in Brazil, are relevant components of **Solid Waste Systems**.

The **ERASMUS+ EGALITARIAN Project** aims to explore the potential for adapting the **Dutch R10 Model** to create a version suitable for developing countries, specifically targeting contexts like Brazil and other developing countries. Initially, the project proposes the development of a modified model, named the **P3RD Model**: **P**= Prevention; **R1**= Reduction; **R2**= Reuse; **R3**= Recycling; and **D**= Disposal. Besides considering that both, the Solid Waste Disposal into Biodigester Sanitary Landfills and the social-economical inclusion of the waste pickers now making a living in the existing open dumpsites, are relevant elements, this modification simplifies the original framework by focusing on five titles, rather than the ten utilized in the **Dutch R10 Model**. This adaptation seeks to establish circular economy project management guides tailored to the needs and capacities of developing nations. It is worthwhile to emphasize that all **Rs** of the **Dutch R10 Model** are included in the five titles of the **P3RD Model**: (i) **Prevention and Reduction via Product Engineering** (**PR1**) include **Refuse**, **Rethink**, **Reduce**; (ii) **Reuse** (R2) encompasses **Reuse**, **Repair**, **Refurbish**, **Remanufacture**, **Repurpose**; (iii) **Recycling** (**R3**) that includes only **Recycling**; and (iv) **Disposal** (D) into **Biodigester Sanitary Landfills**, built in such a way to avoid land contamination and allowing industrial use of the methane gas generated.

As shown in **Figure 1**, the process used by **Cooperatives of Waste Pickers** in Brasilia, for both, **R2-Reuse** and **R3-Recycling** can be described in terms of **Selective Collection→C1**, **Triage→T2**, **Transformation→T2** (**Material**: Biological, Physical, Chemical; and **Organizational**: Environmental, Cooperativism and Education; Decent Work Conditions; Creation of Cooperatives); and **Commercialization→C2**.

Regarding **Environmental Education** in **Organizational Transformation**, a basic premise is that the inclusion and representation of waste pickers in educational and awareness initiatives are relevant issues for fostering sustainable waste management practices. A workshop was then held to develop AI-generated personas to be used in an **App** that authentically represent waste pickers and serve as educators for the external community on proper waste separation and collection. The workshop involved ten waste pickers (seven women and three men) belonging to CENTCOOP, an entity that represents more than twenty cooperatives of waste pickers, jointly conducted by students from AAU, Saxion University and UnB, under the supervision of UnB professors. Using an AI-powered program developed for Educado, an Egalitarian Erasmus+ Project, iteratively generated images based on their input, collecting feedback until they were satisfied with the final personas, one female and one male. Concurrently, participants identified their main struggles in waste collection, reaching a consensus on contamination as the biggest challenge. They elaborated on specific problematic materials, including hospital waste, low-quality recyclables, organic matter, and even animal corpses. Additionally, they shared their expectations for systemic improvements, highlighting the need for fines, external education, and enhanced awareness campaigns. Notably, they requested AI-generated images of children to serve as influencers in educating younger generations. The results demonstrated a strong sense of ownership and engagement among the participants, who expressed satisfaction with the personas, often comparing them to real individuals within

their cooperatives. This study highlights the potential of AI-driven participatory design in empowering marginalized communities, enhancing their visibility, and supporting their advocacy efforts. The findings also underscore the importance of addressing contamination and improving educational strategies to foster a more effective and humane waste management system. Regarding contents of the App, to be developed during the first semester of 2025, a major reference is the **10R Dutch circular economy model**. Considering the scope of the cooperatives, only **R_2-Reuse** and **R_3-Recycling** components of the P3RD model are subjects to be considered in the contents of the App, both encompassing Selective Collection, Triage, Transformation, and Commercialization

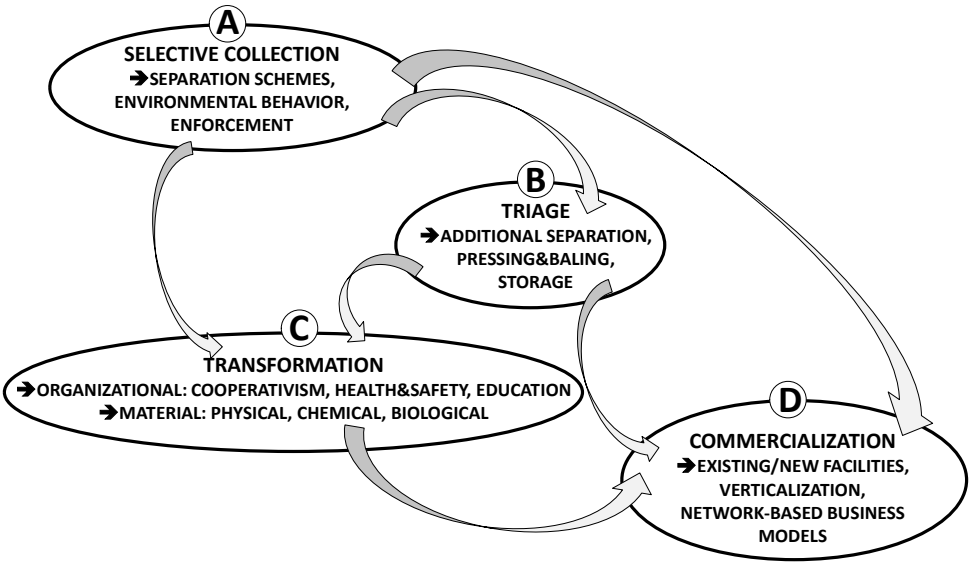


FIGURE 1- REUSE AND RECYCLING C1T1T2C2 PROCESS USED BY COOPERATIVES OF WASTE PICKERS

Taking the **P3RD Model** (an adapted version of the **Dutch R10 Model** for developing countries) as reference, the IEPR's project attention is driven to the applications of the CONFEA EPR Sectors (Megatheme1: 1.3.21 to 1.3.26) related to the components of solid waste systems: **Products P-Prevention&R1-Reduction**, **R2-Reuse**, **R3-Recycling**, and **D-Disposal**.

Focusing on the **P3RD Model**, the first tailoring step is the iterative **Description** of an **Open System** bearing characteristics of **S-Sustainability**, **I-Innovation**, **P-Privacy**, and orientation to **A-Business Analysis adjusted to Risk** (acronym **SIPA**). According to Guarajedagui (Guarajedagui, 2006, pp. 110):

> "Seeing the whole requires understanding Structure, Function, Process, and Context, at the same time.
>
> … Structure defines components and their relationships; function defines the outcomes or results produced; process explicitly defines the sequence of activities, and the know-how required to produce the outcome; context defines the unique environment in which the system is situated."

Figure 2, adaptation of Figure 7.2 (Guarajedagui, 2011, pp. 112), depicts the first generic step of the iterative **Description of an Open System** bearing characteristics of **S-Sustainability**, **I-Innovation**, **P-Privacy**, and orientation to **A-Business Analysis adjusted to Risk** (acronym **SIPA**), for example the **Solid Waste System Components**, in terms of **F-Function**, **S-Structure**, **P-Process** and **C-Context**.

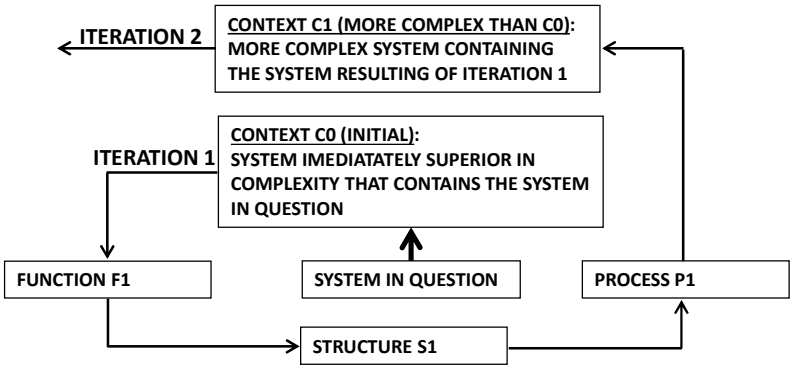


FIGURE 2- OPEN SYSTEM ITERATIVE DESCRIPTION

CONTENT INTEGRATION WORKSHOP-CIW

Focusing on teamwork, the project students are divided into groups of 3 to 5 participants. Initially, each group describes the respective **P3RD Model** component assigned to it, in terms of **F-Function**, **S-Structure**, **P-Process**, and **C-Context**. Such descriptions take into account **S**-Sustainability, **I**-Innovation, **P**-Privacy Protection, and **A**-Business Analysis with Decisions Adjusted to Risk (acronym **SIPA**), according to the **P3RD Model** component assigned to each group: Products oriented to **P**-Prevention and **R1**-Reduction→**SRSPR1.SIPA**, **R2**-Reuse→**SRSR2.SIPA-CC** via Cooperatives of Waste Pickers; **R3**-Recycling→**SRSR3.SIPA-CC** via Cooperatives of Waste Pickers, and **D**-Disposal→**SRSD.SIPA**.

Through teamwork, now with a focus on the description obtained, each group looks for applications of the CONFEA EPR Sector related to the group corresponding P3R component.

As the final delivery, the groups prepare and present a closing tele-event, emphasizing the outcomes as a result of the participation of all students involved as one team.

BRAZILIAN TAX INCENTIVE FOR REUSABLE AND RECYCLABLE MATERIALS

The **Decree No. 12.106**, of **July 10,2024**, that regulates the **Law No. 14.260**, of **December 8, 2021**, is an example of public policy regarding **Circular Economy in Brazil**, with a focus on tax incentives associated with reusable and recyclable materials. It follows a free translation from Portuguese to English of the **Decree No. 12.106**.

"Article 1 The tax incentive for the recycling production chain established in Law No. 14,260, of December 8, 2021, is regulated, with a view to promoting the use of raw materials and inputs from recyclable and recycled materials.

Article 2 Individuals and legal entities taxed on the basis of actual profit may deduct part of the income tax due to direct support to projects previously approved by the Ministry of Environment and Climate Change, aimed at:

I - *capacitation, training and technical advice, including for the promotion of national and international exchanges, for the school and academic areas, business, community associations and social organizations that explicitly include the promotion, development, execution or support of recycling or reuse activities for materials,*

II - *incubation of micro-enterprises, small enterprises, cooperatives and solidarity social enterprises that operate in recycling activities,*

III - *research and studies to support actions that involve shared responsibility for the life cycle of products,*

IV - *implementation and adaptation of physical infrastructure of micro-enterprises, small enterprises, industries, cooperatives and associations of collectors of reusable and recyclable materials,*

V - *acquisition of equipment and vehicles for the selective collection, reuse, processing, treatment and recycling of materials by industries, micro-enterprises, small enterprises, cooperatives and associations of collectors of reusable and recyclable materials,*

VI - *organization and support for marketing networks and production chains, integrated by micro-enterprises, small enterprises, cooperatives and associations of collectors of reusable and recyclable materials,*

VII - *strengthening the participation of collectors of reusable and recyclable materials in recycling chains; and*

VIII - *Development of new technologies to add value to the work of collecting reusable and recyclable materials.*

Article 3 - The deduction of income tax referred to in this Decree shall observe the following limits and conditions:

I - *in relation to individuals, limited to six percent (6%) of the income tax due calculated in the Annual Income Tax Return, as provided for in article 4, caput, item I, of Law No. 14,260, of December 8, 2021; and*

II - *in relation to legal entities, limited to 1% (one percent) of the tax due in each quarterly or annual assessment period, as provided for in article 4, caput, item II, of Law No. 14,260, of December 8, 2021.*

Sole Paragraph. Legal entities may not deduct the amount referred to in the caput for the purposes of determining the actual profit and the calculation basis of the Social Contribution on Net Income.

Article 4 - The administrative procedures related to the presentation, reception, analysis, approval, monitoring, evaluation of results and rendering of accounts shall be established in an act of the Minister of State for the Environment and Climate Change.

Article 5 - The resources from incentives made pursuant to the provisions of this Decree shall be deposited and operated in a specific bank account, opened in a financial institution accredited by the Ministry of Environment and Climate Change, which is held by the proponent of the approved project, and the respective rendering of accounts shall be established in an act of the Minister of State for the Environment and Climate Change.

Article 6 - The Ministry of Environment and Climate Change shall inform the Special Secretariat of the Federal Revenue of Brazil of the Ministry of Finance of the amounts corresponding to the incentives intended for direct support to projects in the previous calendar year.

Article 7 - Act of the Minister of State for the Environment and Climate Change shall regulate the operation of the

National Commission for Recycling Incentive – CNIR, referred to in article 14 of Law No. 14,260, of December 8, 2021.

Article 8 - All information regarding the proposals and projects submitted under Law No. 14,260, of December 8, 2021, is public and will be disclosed in official systems of the Ministry of Environment and Climate Change.

Sole Paragraph. The proposals admitted and eligible for fundraising, in addition to being published in the Official Gazette of the Union, will be made available on the website of the Ministry of Environment and Climate Change, and will contain the following information:

I - *corporate name and National Registry of Legal Entities – CPNJ of the proponent,*

II - *Proposal number and project name,*

III - *object of the proposal,*

IV - *bank account number of the fundraising; and*

V - *Period foreseen for fundraising and project execution. Art. 9 This Decree enters into force on the date of its publication.*

Brasilia, July 10, 2024; 203rd anniversary of Independence and 136th anniversary of the Republic.

LUIZ INÁCIO LULA DA SILVA, Fernando Haddad, Maria Osmarina Marina da Silva Vaz de Lima"

EXPECTED OUTCOME

The expected outcome aims initially at understanding the P3RD system, showing that by describing its components in terms of Function/Structure/Process/Context. Having the corresponding description as reference, teamwork then identifies and characterizes applications and/or projects, focusing on the local communities as primary beneficiaries, which will experience improved resource efficiency, reduced environmental impact, and enhanced economic opportunities. Additionally, the project teamwork is expected to contribute to global knowledge on circular economy practices, offering a model that can be adapted and replicated in other developing regions. Through collaboration with international partners and stakeholders, the **ERASMUS+ EGALITARIAN Project** will expand its global exposure, positioning itself as a valuable tool in the advancement of sustainable development solutions for emerging economies.

By promoting systemic excellence (efficiency/efficacy/eminence/effectiveness), sustainable consumption, addressing climate action, fostering decent work, good health and poverty reduction and other sustainability aspects, the **ERASMUS+ EGALITARIAN Project's P3RD Model** strongly supports the **United Nations Sustainable Development Goals** (**SDGs**). Through these efforts, the **P3RD Model** not only addresses immediate environmental and economic challenges but also contributes to the long-term resilience and sustainability of communities in developing countries.

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